

A Quantitative Theory of Modern Motherhood

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Society for Economics Dynamics

Motivation - Facts from US Current Population Survey

1. Modern mothers significantly more likely to be unmarried:
 - ▶ 12% of mothers unmarried in 1970
 - ▶ 28% of mothers unmarried in 2023

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1. Modern mothers significantly more likely to be unmarried:
 - ▶ 12% of mothers unmarried in 1970
 - ▶ 28% of mothers unmarried in 2023
2. Unmarried mothers typically poor relative to married parents:
 - ▶ $\frac{1}{3}$ of average earnings
 - ▶ 5x more likely to be in poverty

This Paper

- **Questions:**

1. Why did unmarried motherhood increase over time?
2. How did this affect household income/consumption inequality?
3. Long-run goal: How should family structure shape tax & transfer policies?

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- **Approach:**

1. Document facts about motherhood
2. Novel lifecycle theory of relationship and family formation
3. Intuition from simple model; calibrated quantitative model

Outline

- 1 Facts
- 2 Simple Two-Period Model
- 3 Results from Two-Period Model
- 4 Sneak Peek of Lifecycle Model Results

- **NLSY79 and NLSY97 panels**

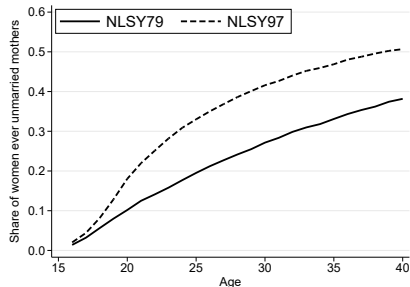
- ▶ NLSY79 cohort born 1957-1964; NLSY97 cohort born 1980-1984
- ▶ Use relationship and fertility history up to age 40 in both

- **National Survey of Family Growth (NSFG)**

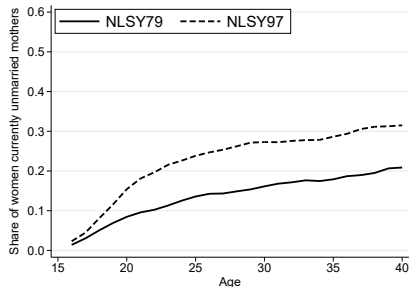
- ▶ Cross-sectional surveys, 12 covering 1973-2023
- ▶ Data on relationships, sex/pregnancy, birth control

Fact 1: Unmarried motherhood is widespread and growing

Prevalence of unmarried motherhood by age



(a) Share ever an unmarried mother

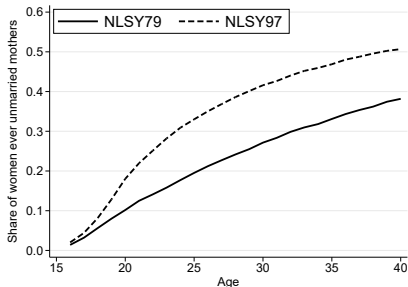


(b) Share currently an unmarried mother

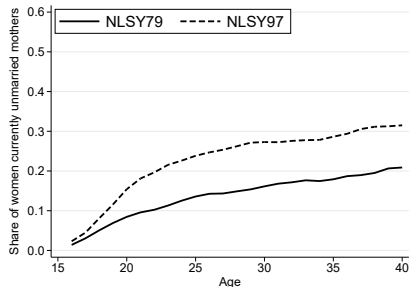
- Share of women **ever** unmarried mother by 40: ↑ from 38% to 51%

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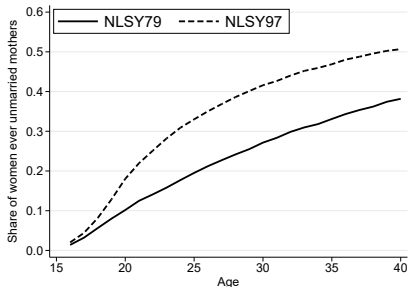


(b) Share currently an unmarried mother

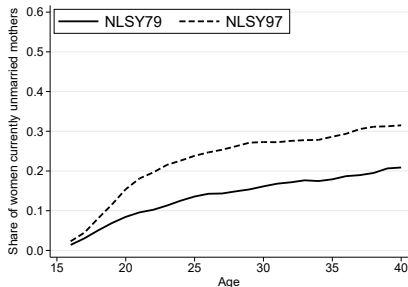
- Share **currently** an unmarried mother $\uparrow \sim 10\text{pp}$ throughout 20s and 30s

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(a) Share ever an unmarried mother



(b) Share currently an unmarried mother

- NLSY97: 63% of **mothers** ever unmarried; 41% of motherhood-years.

Fact 2: Most Mothers Now Give Birth Before Marriage

Relationship at first birth and marriage rates

Relationship at first birth	Share of all mothers (%)		Share married by 40 (%)	
	NLSY79	NLSY97	NLSY79	NLSY97
Single	20.3	30.6	63.4	54.8
Cohabiting	4.0	21.6	71.2	60.0
Total	24.3	52.2	64.7	57.0

- Share of mothers unmarried at first birth **increased** from $\sim \frac{1}{4}$ to $> \frac{1}{2}$

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- Moms not just delaying marriage; share ever married by 40 **decreased**.

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- Details:

[More on cohabitation vs marriage](#)[More on cohabiting at first birth](#)

Fact 3: Most Pre-Marital Births Are Unintended

Share of live births resulting from unintended pregnancy

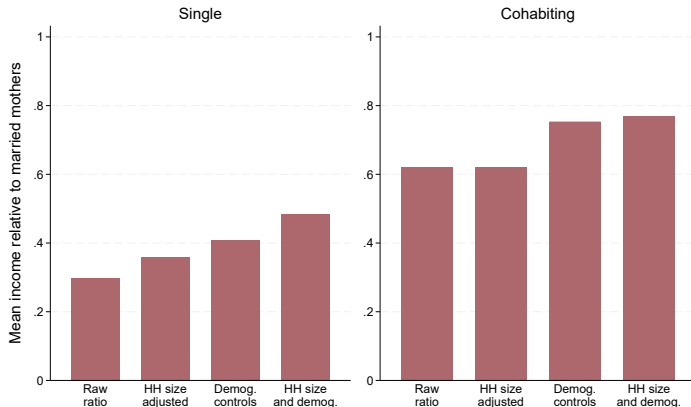
Relationship at first birth	NLSY79 cohort (born 1957–1964)	NLSY97 cohort (born 1980–1984)
Single	55.7	66.9
Cohabiting	47.0	47.0
Married	23.3	22.8

Source: NSFG surveys 2002–2023; mothers who were 16–39 years old at time of birth.

- For mothers single/cohabiting at birth, $\frac{1}{2}$ to $\frac{2}{3}$ of births are unintended
- *Ex post* survey; likely **underestimates** *ex ante* unintended shares

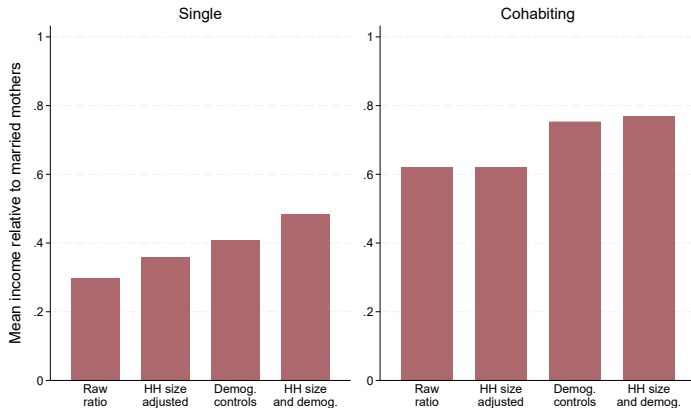
(Rackin & Morgan, 2018; Ralph, Foster & Rocca, 2020)

Fact 4: Unmarried-Mother Households Are Typically Poor



- Single mothers: avg income 30% of married couples
- After adjusting for HH size and demographics, still only 48% [More](#)

Fact 4: Unmarried-Mother Households Are Typically Poor



- Cohabiting mothers: avg income 62% of married couples
- After adjusting for HH size and demographics, 77%. [More](#)

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3. Most pre-marital births are unintended
⇒ sex provides utility but also entails risk; marriage acts as insurance
4. Unmarried-mother households are typically poor
⇒ need to understand why to study policy & welfare implications

Outline

- ① Facts
- ② Simple Two-Period Model
- ③ Results from Two-Period Model
- ④ Sneak Peek of Lifecycle Model Results

Timing, Demographics, and Endowments

- Two periods
- Model populated by:
 - ▶ Unit mass of females (φ) indexed by $i \in [0, 1]$
 - ▶ Unit mass of males (σ) indexed by $j \in [0, 1]$
- Exogenous income: $y_i = y^\varphi \forall i$ and $y_j = y^\sigma \forall j$
- No taxes or transfers

Couples

- At start of first period, each female draws a male
 - ▶ Potential couple defined by state vector (y_i, y_j, b)
- Partnership bliss, b , is AR-1: $b' = \rho b + \epsilon; \quad \epsilon \sim N(\mu_b, \sigma_b)$
- Each couple makes a partnership choice for the remainder of the period:
 - ▶ $p' = s$: separate, draw new partner next period
 - ▶ $p' = u$: unmarried couple
 - ▶ $p' = m$: married couple
- Marriage is a commitment device: divorce costs each partner utility δ

The Birds and the Bees

- All individuals born childless, $n = 0$
- Couples that do not separate then make a sex/fertility choice:
 - ▶ $f \in [0, \bar{f}]$, where $\bar{f}$ is the probability of having a child, $n' = 1$
 - ▶ Each partner receives flow utility $\alpha(f) \geq 0$, strictly increasing in f

Utility from Children

- In couples, both parents receive non-rival utility ν each period
- If parents separate, child stays with mother
 - ▶ Separated mothers continue to receive utility ν each period
 - ▶ Separated fathers no longer receive utility from child

Preferences: Singles

Period utility additively separable over time with discount rate β :

$$w^{\sigma}(c) = \frac{\left(\frac{c}{\sqrt{1}}\right)^{1-1/\omega}}{1 - 1/\omega}$$

$$w^{\varphi}(c, n) = \frac{\left(\frac{c}{\sqrt{1+n}}\right)^{1-1/\omega}}{1 - 1/\omega} + \nu n$$

Difference: If $n = 1$, females bear child costs and receive utility benefit

Preferences: Couples

Period utility additively separable over time with discount rate β :

$$w^{\varphi}(c, f, b, n) = w^{\sigma}(c, f, b, n) = \underbrace{\frac{\left(\frac{c}{\sqrt{2+n}}\right)^{1-1/\omega}}{1-1/\omega}}_{\text{consumption utility}} + \underbrace{\nu n}_{\text{child utility}} + \underbrace{(b + \alpha(f))}_{\text{relationship utility}}$$

Benefit for couples: *split* costs of children; child utility is *non-rival*

How Couples Make Decisions

- Given state (y_i, y_j, b, p, n) , couple chooses relationship $r = (p', f)$
- Timing:
 1. Observe current period state
 2. Woman proposes menu of relationship options
 3. Man selects preferred option
- Details: Formal optimization and value functions

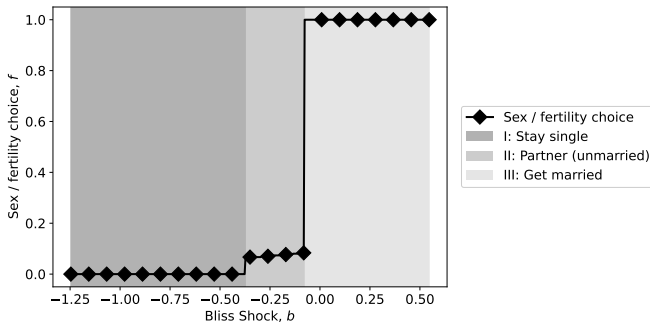
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Working Backward from Period 2

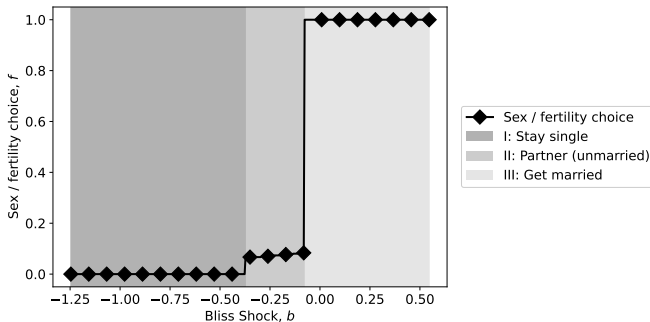
- For now, assume equal incomes: $y^{\varphi} = y^{\sigma}$
- Males:
 - ▶ Weakly dominant to enter last period as unmarried partner with a child
 - ▶ Benefit from consumption sharing; child provides positive net utility
 - ▶ Can separate at no cost in case of bad bliss shock
- Females:
 - ▶ If low b , prefer unmarried and no child
 - ▶ If high b , prefer married with child
 - ▶ Moderate b poses risk of male leaving, marriage provides insurance
- Details: Period 2 value functions

Period 1 Choices with $y^{\varphi} = y^{\sigma}$



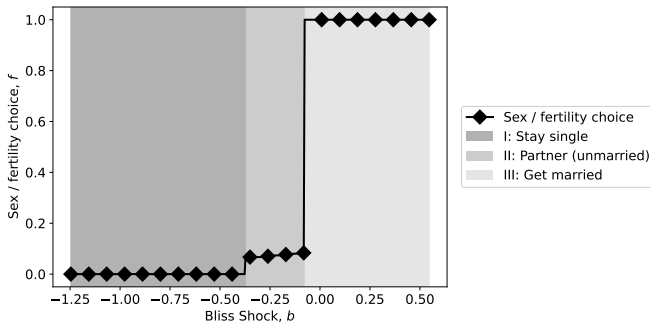
- Region III: High b , couples confident of relationship quality $\Rightarrow$ marry
- Child is net positive $\Rightarrow$ all births in married couples are “intended”

Period 1 Choices with $y^{\varphi} = y^{\sigma}$



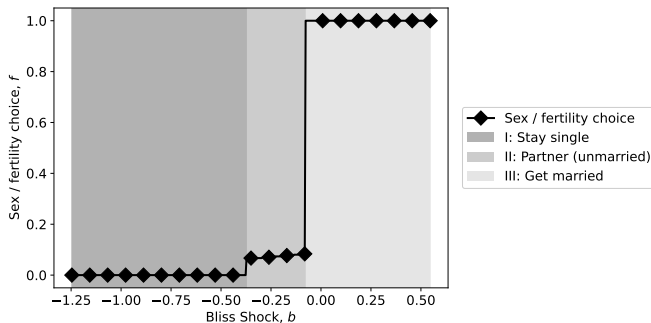
- Region II: Moderate b , greater risk of low $b' \Rightarrow$ unmarried partners
- Choose $f < 1$, increasing in $b \Rightarrow$ some “unintended” births

Period 1 Choices with $y^{\varphi} = y^{\sigma}$



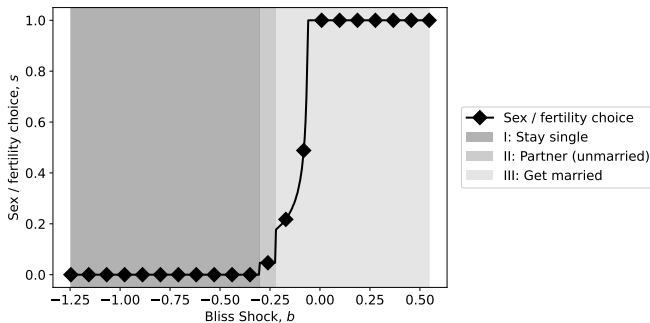
- Region I: Low $b \Rightarrow$ stay single and search again next period
- No consumption sharing and $f = 0$

Period 1 Choices with $y^{\text{♀}} = y^{\text{♂}}$



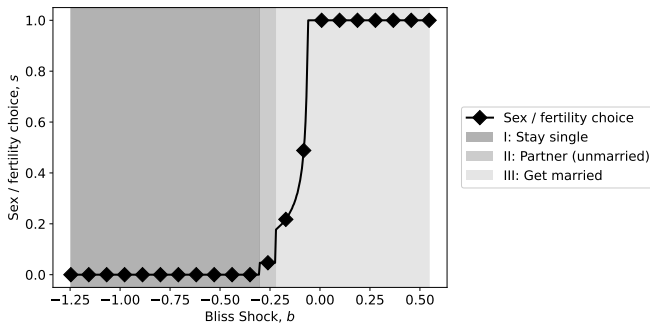
What happens if relative female income is **lower**?

Period 1 Choices with Lower Female Income ($y^{\text{♀}}/y^{\text{♂}} = 0.75$)



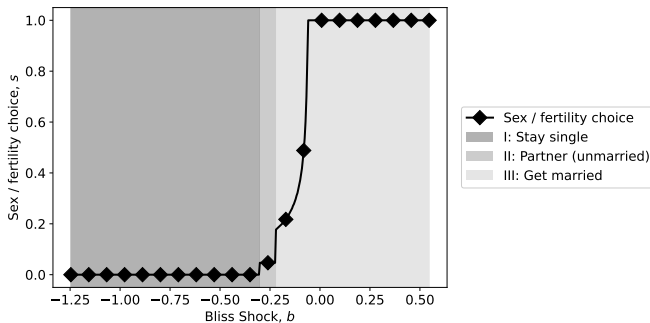
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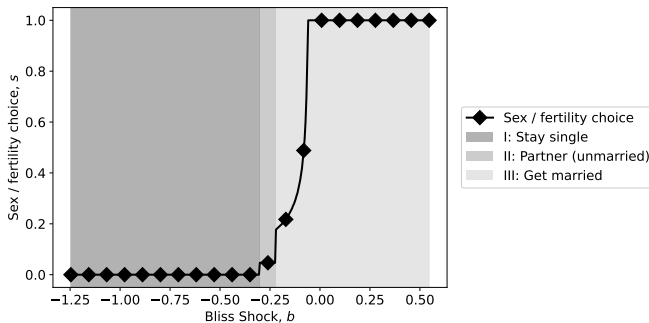
- Region II/III cutoff shifts left $\Rightarrow$ more couples choose marriage
- $p' = u$ is riskier, so females offer higher $f \Rightarrow$ males more willing to marry

Period 1 Choices with Lower Female Income ($y^{\text{♀}}/y^{\text{♂}} = 0.75$)



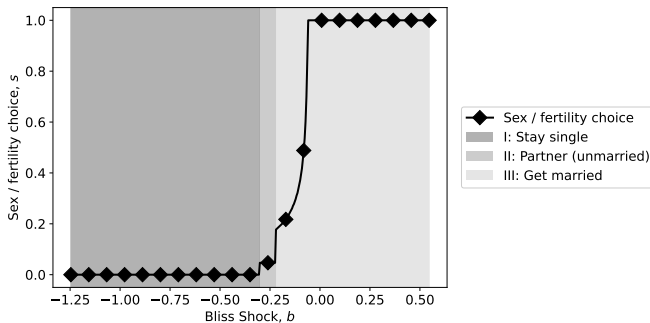
- Low b marriages choose higher f than if they were unmarried partners
- More unmarried moms via divorce, fewer single/partnered at birth

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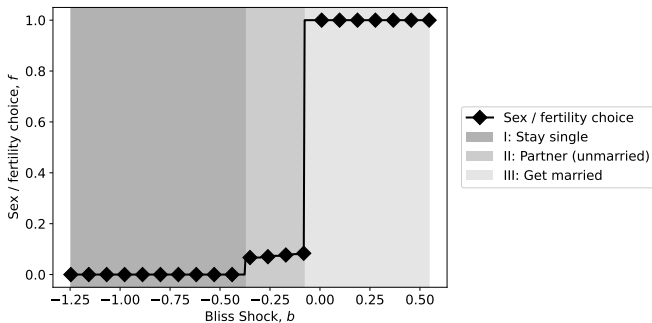
- Region I/II cutoff shifts right $\Rightarrow$ fewer unmarried partners, more singles
- Females offer lower f (riskier); males less willing to partner

Period 1 Choices with Lower Female Income ($y^{\text{♀}}/y^{\text{♂}} = 0.75$)



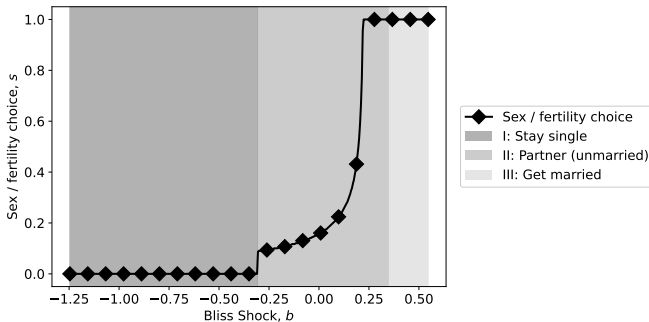
- Testable implications: faster marriages, more divorce
- And fewer unmarried partnerships, fewer unmarried moms at birth

Period 1 Choices with $y^{\varphi} = y^{\sigma}$



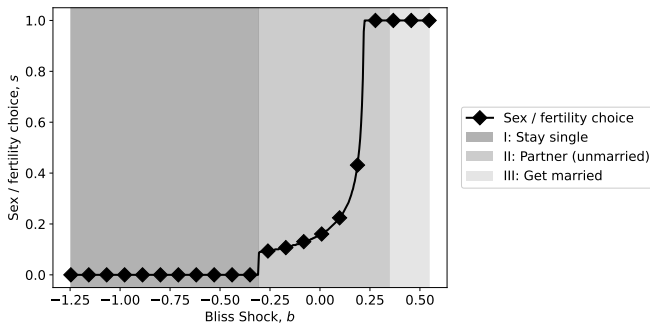
What happens if relative female income is **higher**?

Period 1 Choices with Higher Female Income ($y^{\text{♀}}/y^{\text{♂}} = 1.25$)



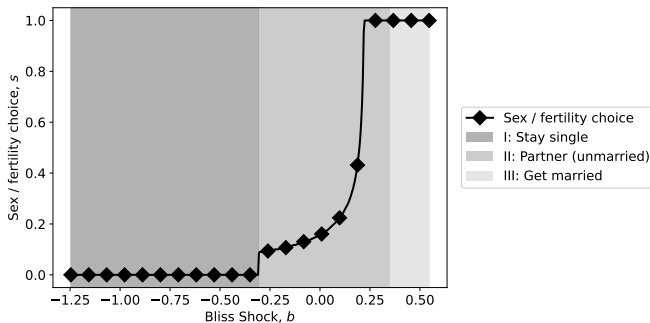
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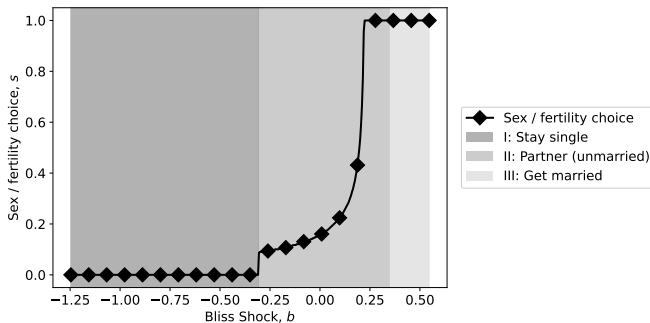
- Region II/III cutoff shifts right $\Rightarrow$ less marriage
- Females: lower cons. sharing benefit, lower risk of single motherhood

Period 1 Choices with Higher Female Income ($y^{\text{♀}}/y^{\text{♂}} = 1.25$)



- Region I/II cutoff also shifts right relative to $y^{\text{♀}} = y^{\text{♂}}$ case
- Now driven by females less willing to tolerate low b

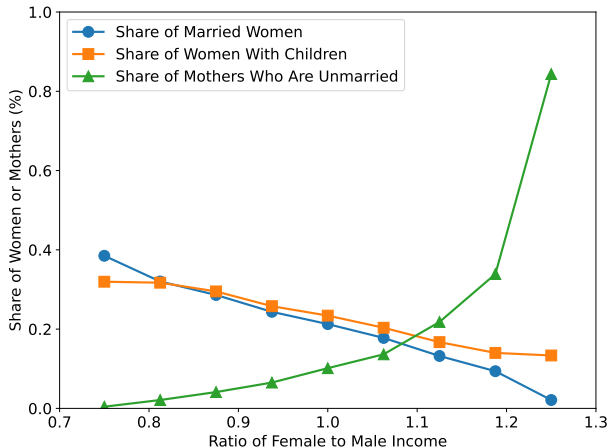
Period 1 Choices with Higher Female Income ($y^{\text{♀}}/y^{\text{♂}} = 1.25$)



- Testable implications: lower marriage rate, delayed marriage
- Lower divorce, more unmarried partners at birth

Summary of Comparative Statics in Two-Period Model

Marriage and Motherhood as a Function of y^{φ}/y^{σ}



Key Takeaways from Two-Period Model

1. y^{φ}/y^{σ} is key for relationship and fertility decisions
 - ▶ Marriage and fertility **decrease** as female income increases
 - ▶ Unmarried motherhood **increases** as female income increases
2. Additional implications as y^{φ}/y^{σ} increases:
 - ▶ Longer “dating” phase, higher quality marriages, lower divorce risk
 - ▶ Fewer unmarried mothers from divorce, more single/cohabiting mothers

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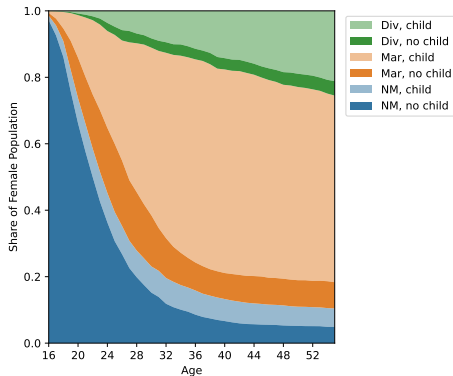
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Towards a Calibrated Lifecycle Version of the Model

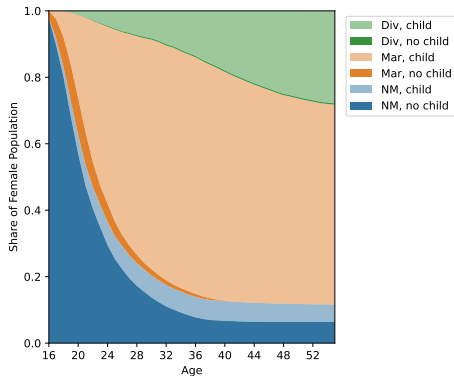
- Results from earlier version:
 - ▶ No income inequality
 - ▶ No government taxes/transfers
 - ▶ Rough calibration to NLSY79 data
- Proof of concept that we can simultaneously generate:
 - ▶ Cross-sectional shares of family types in the data
 - ▶ Accurate lifecycle transition rates between relationship/fertility states

Family Shares

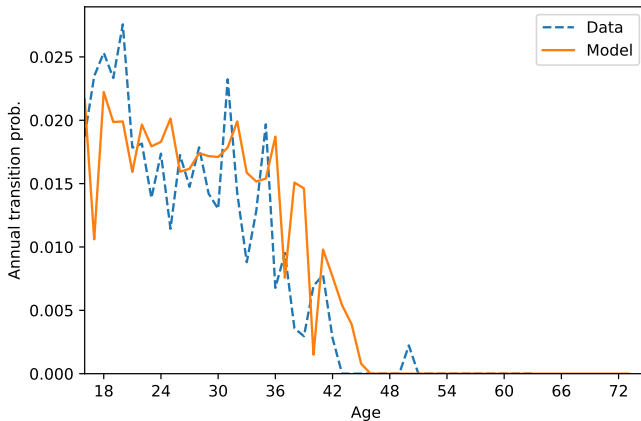
NLSY79



Model

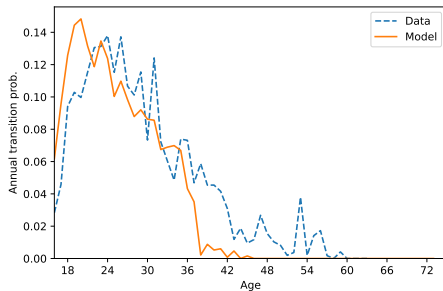


Transitions: Pre-Marital Births for Never-Married Women

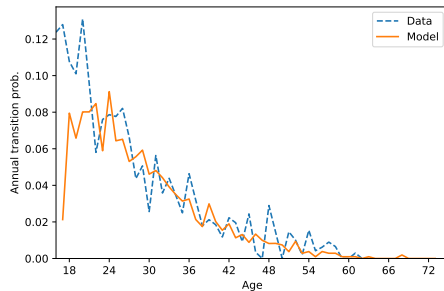


Transition Rate Into Marriage

Never-Married, No Child

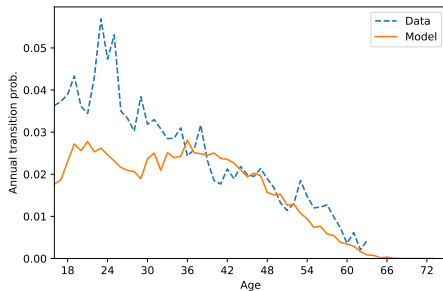


Never-Married, Child

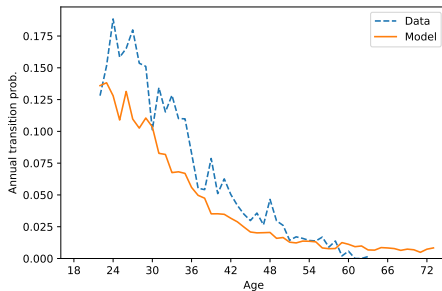


Transition Rates: Divorce and Re-Marriage

Divorce Rate



Re-Marriage Rate



Conclusions

- Document new facts on modern motherhood focused on the **increasing prevalence and changing nature** of unmarried motherhood
- Novel lifecycle theory of relationship and family formation
 - ▶ Endogenous marriage, fertility, divorce, and remarriage
 - ▶ Unmarried motherhood as single, cohabiting, and divorced
- Work in progress: use the model as a laboratory
 - ▶ Disentangle forces causing rise of unmarried motherhood
 - ▶ Understand how changing family structure affects household inequality
 - ▶ Study role of tax and transfer policies with endogenous family types

APPENDIX

Why are these questions non-trivial? Why do we need a model?

- Answers require constructing a counterfactual world in which unmarried motherhood is less common. Is this...
 - ▶ Because fewer women have children?
 - ▶ Because more women marry or fewer divorce?
 - ▶ Because the timing of marriage and childbirth changes?

Why are these questions non-trivial? Why do we need a model?

- Answers require constructing a counterfactual world in which unmarried motherhood is less common. Is this...
 - ▶ Because fewer women have children?
 - ▶ Because more women marry or fewer divorce?
 - ▶ Because the timing of marriage and childbirth changes?
- In each case, we must then ask...
 - ▶ Which individuals change their behavior?
 - ▶ Why do they make different decisions?
 - ▶ What are the economic and policy implications?

Fact 1: Unmarried motherhood is widespread and growing

Relationship history of mothers: NLSY79 and NLSY97

Relationship state	% of mothers ever in relationship state		% of motherhood years before age 40	
	NLSY79	NLSY97	NLSY79	NLSY97
Married	91.3	77.6	68.7	59.4
Unmarried	49.6	63.2	31.3	40.6
Single	47.3	54.5	25.9	26.2
Cohabiting	22.1	49.9	5.4	14.4

- Marriage ↓, and share of mothers ever unmarried ↑ from $\frac{1}{2}$ to nearly $\frac{2}{3}$

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- Share of motherhood years spent married ↓ from 69% to < 60%

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- Share of motherhood years spent unmarried ↑ from 31% to > 40%

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- Most unmarried mothers spend time both single and cohabiting, but cohabitation ↑ more from 79 to 97 cohort

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- Yet single time is still $\sim \frac{2}{3}$ of unmarried motherhood, cohabitation is $\frac{1}{3}$

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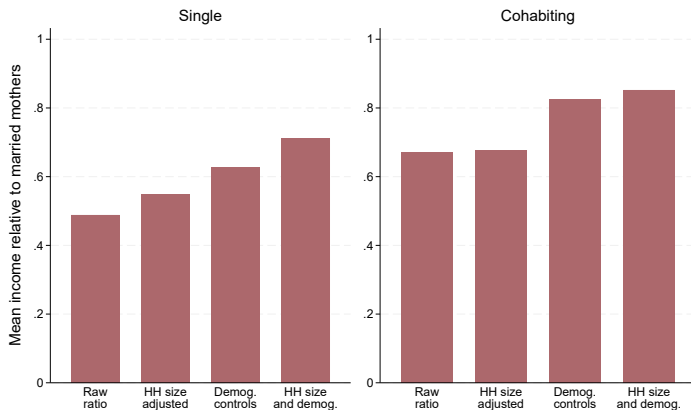
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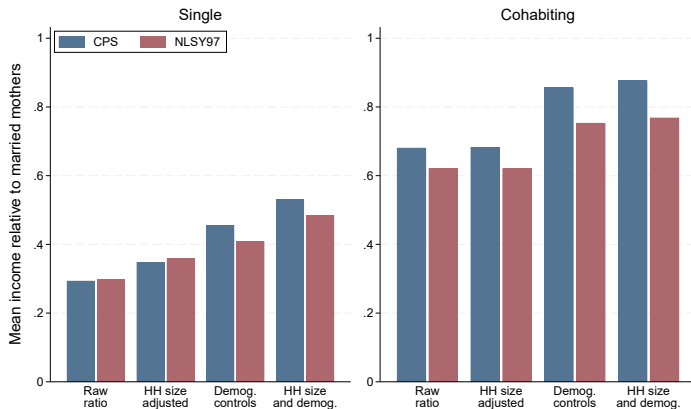
Fact 4 Appendix

- Graph: NSLY97 alternative using family income
- Graph: comparison with CPS using individual + spouse income
- Graph: comparison with CPS using family/household income
- Regression: NLSY97 using individual + spouse income
- Regression: NLSY97 alternative using family income
- [Back to Fact 4](#)

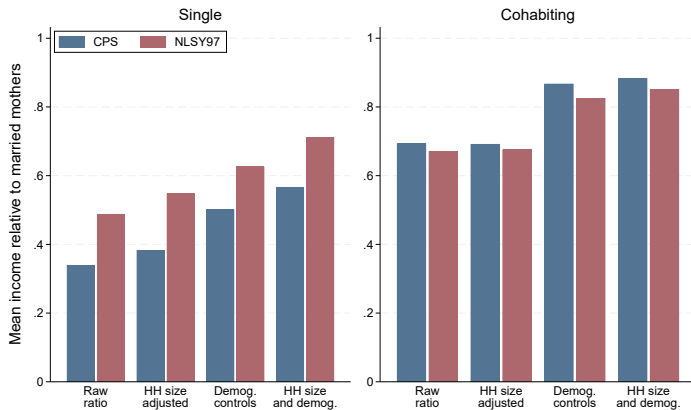
Fact 4 Appendix: NLSY97 Alt Using Family Income



Fact 4 Appendix: NLSY97 vs CPS with Individual Income



Fact 4 Appendix: NLSY97 vs CPS with Family/HH Income



Fact 4 Appendix: NLSY97 Regression with Ind. Income

	ln(income)	ln(OECD-adjusted income)
Single mother	-1.105***	-0.914***
Cohabiting mother	-0.384***	-0.368***
Education (omitted: high school)		
Less than HS	-0.393***	-0.454***
Some college	0.121***	0.120***
College plus	0.506***	0.540***
Implied ratio to married mothers: exp(coefficient)		
Single mother	0.331	0.401
Cohabiting mother	0.681	0.692
Demographic controls	Yes	Yes
Survey years	2010–2019	2010–2019
Observations	10,539	10,539
R-squared	0.457	0.405

Notes: Reference category married mothers with high school education. Regressions are weighted and include demographic controls for race, mother's age, mother's age squared. Observation counts are unweighted respondent-year counts.

Fact 4 Appendix: NLSY97 Regression with Family Income

	ln(income)	ln(OECD-adjusted income)
Single mother	-0.661***	-0.499***
Cohabiting mother	-0.281***	-0.245***
Education (omitted: high school)		
Less than HS	-0.401***	-0.432***
Some college	0.147***	0.165***
College plus	0.512***	0.562***
Implied ratio to married mothers: exp(coefficient)		
Single mother	0.516	0.607
Cohabiting mother	0.755	0.783
Demographic controls	Yes	Yes
Survey years	2010–2019	2010–2019
Observations	12,105	12,105
R-squared	0.314	0.293

Notes: Reference category married mothers with high school education. Regressions are weighted and include demographic controls for race, mother's age, mother's age squared. Observation counts are unweighted respondent-year counts.

Mean number and duration of relationships through age 40

	Cohabitation	Marriage
<u>All relationships</u>		
Mean number of relationships	1.83	1.20
Mean duration (years)	2.50	9.92
<u>Relationships with child < 18 present</u>		
Mean number of relationships	1.68	1.16
Mean duration (years)	3.15	10.63

Notes: Statistics are conditional on being in a cohabiting relationship or marriage (exc. zeros).

- Relationships through age 40 of mothers **cohabiting** at first birth:
 - ▶ 53% separate from that partner
 - ▶ 34% marry that partner
 - ▶ 14% remain in cohabiting relationship with that partner

- Attachment Theory:
 - ▶ **Attachment** is a bonding system between caregivers and children
 - ▶ Roots in evolutionary biology and developmental psychology
 - ▶ Generates emotional connection and facilitates child survival
- Evidence for Mothers:
 - ▶ Stronger biological priming (pregnancy, lactation, oxytocin/dopamine)
 - ▶ Attachment tightly linked to responsiveness to child needs
- Evidence for Fathers:
 - ▶ Weaker biological priming
 - ▶ Developed through time investment, engagement, and play

Model Simplifications to Build Intuition

- Two periods
- Log consumption
- No taxes or transfers ($T(y; m', n) = 0$)
- No income heterogeneity ($y_i = y^{\varphi} \forall i$ and $y_j = y^{\sigma} \forall j$)
- Deterministic child attachment ($\phi(p') = 1 \forall p'$)
- Details: Parameters

State Space

- An individual's state is a vector:
 - ▶ $t = 1, \dots, T$ — age
 - ▶ $n = 0, 1$ — children
 - ▶ $a = 0, 1$, — male attachment to child (if present)
 - ▶ p — partnership status
 - ▶ b — bliss shock
 - ▶ y_i or y_j — income endowment
- To keep notation clean, we define a couple's state by:
 - ▶ $x = (n, a, p, b, y_i, y_j)$

Optimization Problem

- Let $V_t^k(x)$ be the individual value function for sex $k \in \{\varphi, \sigma\}$ in state x
- The conditional value function for relationship $r = (p', s)$ is given by:

$$\begin{aligned}\tilde{V}_t^k(r_s; x) &= w^k(r_s; x) - \delta \mathbb{1}_{m=1} + \beta \mathbb{E}_{x'} \left[V_{t+1}^k(x') \mid p' = s \right] \\ \tilde{V}_t^k(r \neq r_s; x) &= w^k(r; x) + \beta \mathbb{E}_{x'} \left[V_{t+1}^k(x') \mid p' \neq s \right]\end{aligned}$$

where the expectations for each sex are defined in the following slides...

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Expected Value if Couple Stays Together

$$\underbrace{\mathbb{E}_{x'} \left[V_{t+1}^k(x) \mid p' \neq s \right]}_{\text{couple stays together}} = \int_{b'} \left(\underbrace{(1-f)(1-n)}_{\text{probability: no child}} V_{t+1}^k(0, 0, p', b', y_i, y_j) \right. \\ + \underbrace{[f(1-\phi(p'))(1-n) + (1-a)(1-\phi)n]}_{\text{probability: child, man unattached}} V_{t+1}^k(1, 0, p', b', y_i, y_j) \\ \left. + \underbrace{[f\phi(p')(1-n) + (a + (1-a)\phi)n]}_{\text{probability: child, man attached}} V_{t+1}^k(1, 1, p', b', y_i, y_j) \right) d\psi(b' \mid b)$$

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Expected Value for Women if Couple Separates

$$\underbrace{\mathbb{E}_{x'} \left[V_{t+1}^{\varnothing}(x) \mid p' = s \right]}_{\text{women: couple separates}} = \int_{b'} \int_{y_j} \left(\underbrace{[1 - n]}_{\text{probability: no child}} V_{t+1}^{\varnothing}(0, 0, p', b', y_i, y_j) \right. \\
 + \underbrace{[(1 - \phi)n]}_{\text{probability: child, man unattached}} V_{t+1}^{\varnothing}(1, 0, p', b', y_i, y_j) \\
 \left. + \underbrace{[\phi n]}_{\text{probability: child, man attached}} V_{t+1}^{\varnothing}(1, 1, p', b', y_i, y_j) \right) d\lambda_{t+1}^{\sigma}(y_j) d\psi_0(b')$$

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Expected Value for Men if Couple Separates

$$\underbrace{\mathbb{E}_{x'} \left[V_{t+1}^{\sigma}(x) \mid p' = s \right]}_{\text{men: couple separates}} = \int_{b'} \int_{y_i} \left(\underbrace{[1 - \pi_{t+1}(y_i)]}_{\text{probability: no child}} V_{t+1}^{\sigma}(0, 0, p', b', y_i, y_j) \right. \\ + \underbrace{[(1 - \phi)\pi_{t+1}(y_i)]}_{\text{probability: child, man unattached}} V_{t+1}^{\sigma}(1, 0, p', b', y_i, y_j) \\ \left. + \underbrace{[\phi\pi_{t+1}(y_i)]}_{\text{probability: child, man attached}} V_{t+1}^{\sigma}(1, 1, p', b', y_i, y_j) \right) d\lambda_{t+1}^{\circ}(y_i) d\psi_0(b')$$

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Illustrative Model Parameters

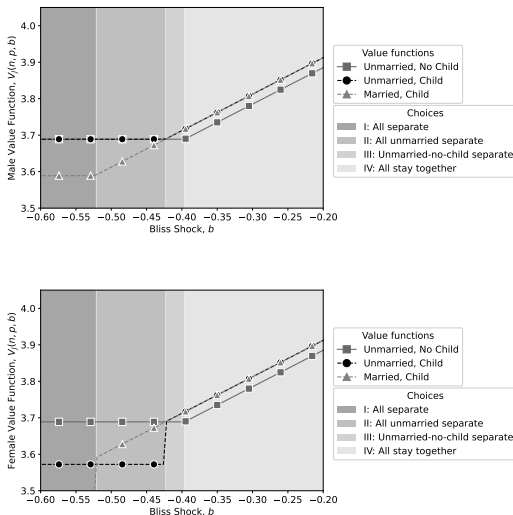
<u>Parameter</u>	<u>Interpretation</u>	<u>Value</u>
μ_b	Mean of bliss process	-0.35
σ_b	SD of bliss innovation	0.32
ρ	Persistence of bliss process	0.70
$\alpha(f)$	Utility from sex (functional form)	$\alpha_0(\frac{f}{f+\alpha_1})$
α_0	Utility from sex	0.05
α_1	Utility from sex	0.01
$\bar{f}$	Maximum fertility probability	1.00
β	Discount factor	0.96
ν	Utility from children	0.23
δ	Divorce cost	0.10

Period 2 Value Functions with $y^{\varphi} = y^{\sigma}$

Region IV:

- All couples stay together
- Net benefit of child > 0
- Value functions increase linearly with bliss, b
- Indifferent between marriage and unmarried partnership (no period 3)

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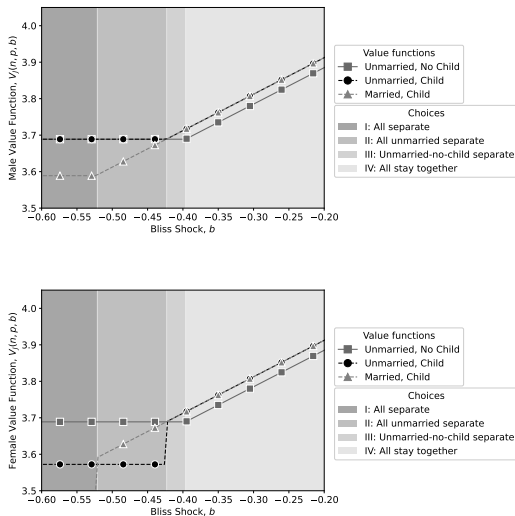


Period 2 Value Functions with $y^{\varphi} = y^{\sigma}$

Region III:

- Unmarried couples without a child separate
- Intuition: utility gain from consumption sharing + sex not enough to overcome low bliss shock
- Value function flat w.r.t. b

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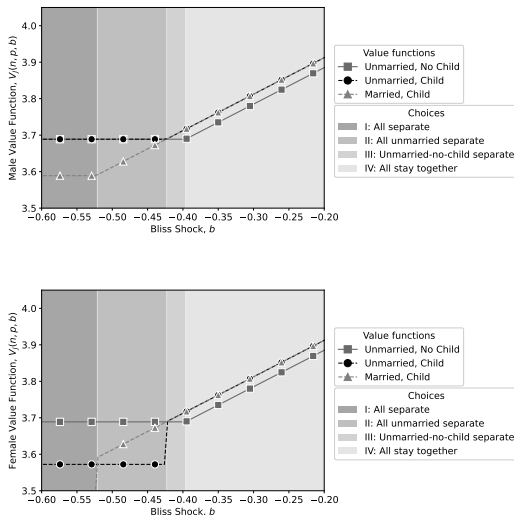


Period 2 Value Functions with $y^{\varphi} = y^{\sigma}$

Region II:

- All unmarried separate
- Males: utility from child no longer large enough to offset low bliss
- Females: would be better off married; value function drops discretely as she bears cost of child alone

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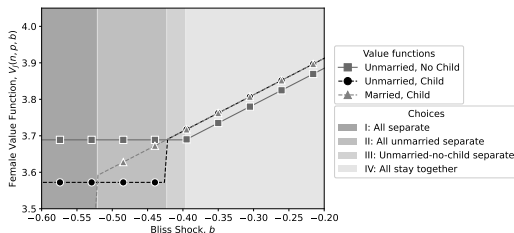
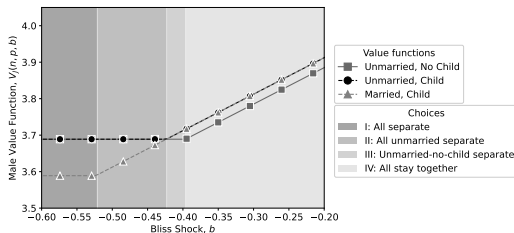


Period 2 Value Functions with $y^{\varphi} = y^{\sigma}$

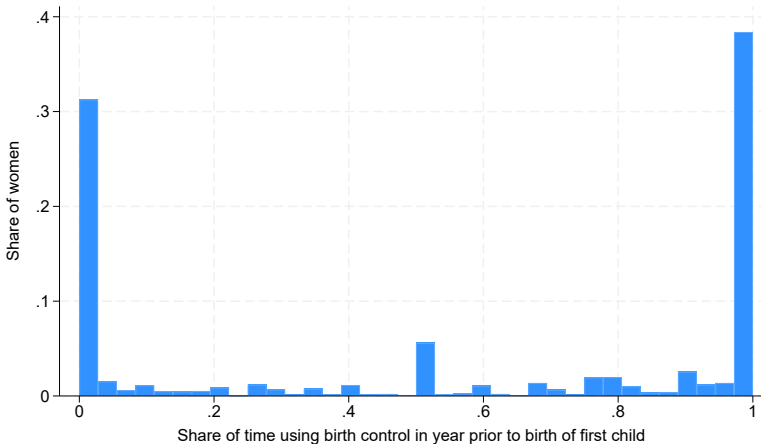
Region I:

- All couples separate
- Bliss shock low enough that married couples now willing to pay divorce cost, δ

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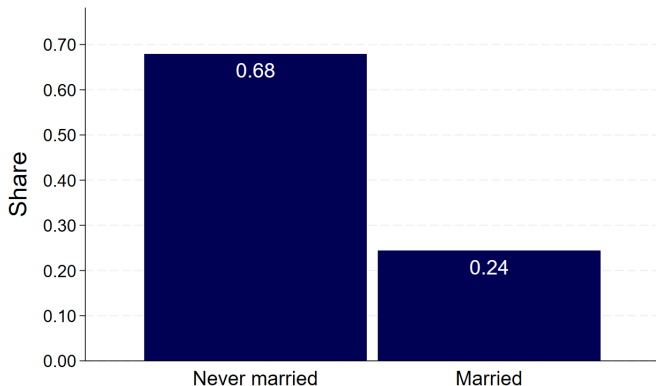


NLSY97: Birth Control Usage Before First Child



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Share of Live Births Among Mothers



Unintended Births by Relationship Status and Education

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Share of Live Births Among Mothers

